Kawya Dhawalindini

Electronics & Power Systems Engineering Undergraduate

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LinkedIn Profile

Engineering Portfolio

PROFILE

Electronics and Power Systems Engineering undergraduate at the Sri Lanka Technological Campus with a strong interest in power generation, electrical power systems, renewable energy integration, and energy storage technologies. MATLAB/Simulink-based modeling of renewable energy and hydrogen energy storage systems, with additional experience in embedded systems and engineering tools. Interested in applying technical knowledge, analytical skills, and problem-solving abilities to practical engineering challenges while gaining industry experience in the power and energy sector.

EDUCATION

Sri Lanka Technological Campus (SLTC) 2022 – 2026
BSc. (Hons) Electronic and Power System Engineering Sri Lanka

Pushpadana Girls' College, Kandy 2013 – 2021
GCE Advanced Level – Physical Science Stream

TECHNICAL AREAS

Power Systems

- Power System Analysis
- Power System Control & Protection
- Smart Electrical Grids
- High Voltage Engineering

Renewable Energy

- Renewable Energy Systems
- Energy Storage
- Green & Sustainable Technologies

ACADEMIC PROJECTS & RESEARCH

AI-Assisted Smart Energy Management System *ongoing*

- Developing a system that monitors energy consumption and applies AI-based analysis to optimise loads.

Statistical Analysis of EV Battery Degradation *R software*

- Conducted statistical analysis of EV battery degradation using R, including data processing, visualization, and interpretation of battery performance trends.

Hydrogen Energy Storage Simulation *MATLAB / Simulink*

- Developed a simulation model for hydrogen-based renewable energy storage, covering hydrogen production, storage, and energy conversion.

Password-Based Smart Door Lock *Embedded Systems*

- Developed a microcontroller-based smart door lock using password authentication for improved residential security.

TECHNICAL SKILLS

Simulation & Modelling: MATLAB, Simulink, PS CAD
Electrical / Engineering Tools: AutoCAD, Proteus
Programming: Python, R, Arduino / Microcontroller Programming
Productivity: Microsoft Word, Excel, PowerPoint

LEADERSHIP & PROFESSIONAL ACTIVITIES

Secretary – Spark Wave Seminar Series 2025
Sri Lanka Technological Campus

Member – IEEE Power & Energy Society (PES) Sri Lanka Chapter 2025 – Present

CERTIFICATIONS

Foundational Certificate Course on ESG Fundamentals	IEEE Y2NPRO – 2026
AWS Cloud Computing Fundamentals	Amazon Web Services – 2025
Climate Change Negotiation	UNFCCC – 2025
Inclusive Growth	International Monetary Fund – 2025
Python Essentials 1	Cisco Networking Academy – 2024

LANGUAGES

Sinhala: Native — **English:** Fluent — **Japanese:** N5 Level

REFERENCES

Dulashi Karunakalage
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